Final Project Design Phase

Computer Science - Week 11

Oct 3, 2026 - Version 0.0.4

For complete instructions and project ideas, refer to the assignment in Canvas. Specifically, review the final project requirements before you continue below. If you have not reviewed the project requirements in Canvas (you might have to go to the modules view and check the current or past week's content), you will not be able to properly complete the parts below. In addition, it may be helpful to review the information on this Drafter documentation page.

0) Project Planning and Preparation

Successful programmers spend time planning before they begin coding. *After* reviewing the project requirements, answer the following questions to help guide your design and development process.

A. What concepts from class or previous assignments might help you complete this project? Which programming concepts (loops, conditionals, lists, classes, etc.) do you expect to use? What programming skills do you want to improve while completing this project?

Reading from a file, helper functions, lists, loops, drafter, classes, basically everything. I want to improve drafter skills and web page building skills

B. Which parts of the project do you expect to be the easiest for you? The hardest? Why?

The actual science part will be the easiest, and coding it wont be too too hard. Mainly, the issue will be making it work universally. I don’t know how I’d work with coefficients and subscripts and etc.

C. Even though you have not yet designed the final project, estimate which tasks will likely require the most time and effort? Which features or components should be implemented first?

All my conversions should be made first as helper functions, then functions for coefficients, then math functions, then the bigger functions that display things on

screen.

D. What distractions might interfere with your work and how will you minimize them? What resources will you need available while working? What questions, if any, do you still have about the assignment requirements?

Social media, other classes, sleep. I'll set time out of my day to work on the project with no distractions

1) Site Diagram

Create a mock-up of your planned website. We suggest you make the mockup on paper, although you can also use software like PowerPoint or Slides to make a mockup. There is other design software that could also be used, but we strongly encourage you to keep it simple. Do not spend more than 15-30 minutes detailing the mock-up. Your goal should be to make a very quick and dirty diagram, to show that you have some kind of idea.

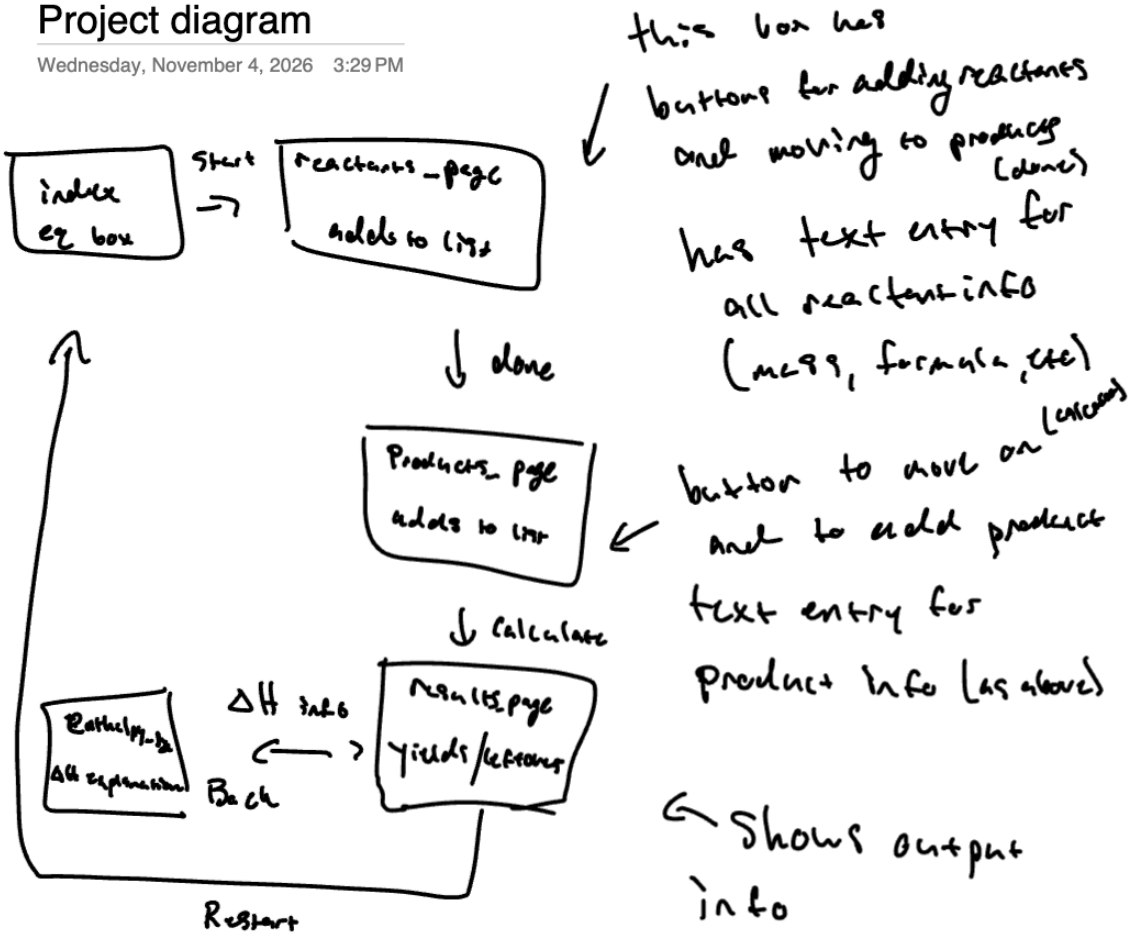
We recommend drawing each page as a rectangle, and laying out the text, buttons, and other components inside. Then, use arrows to indicate links between pages. Label each page the way you will name each function, and then put labels on the arrows to indicate what the button links. Also write down any parameters that should come into the pages.

In addition to the pages and routes, write down what fields your `State` will have, both their names and types. Think clearly about what data will need to be persistent across the entire website.

Add photo(s) or screenshot(s) of your website mock-up and labeled diagram on the next page.

Project diagram

Wednesday, November 4, 2026 3:29 PM



State

balanced_eq : Str

reactants: list [Compound]

products: list [Compound]

lim_reagent : Str

enthelpy : float

exothermic : bool

show_work : bool

message : Str

Compound

formula : Str

coefficient : int

molar_mass : float

mass : float

remaining : float

Yield_theory : float

Yield_percent : float

heat_per_mol : float

2) Implementing Requirements

This table should serve as a draft for how you will implement the requirements into your final project. You may change your ideas later on, but your final site must fulfill all the requirements.

Requirement	**Implementation**
Purpose/Functionality A 1-2 sentence description of the purpose or functionality of your site	**Take a balanced chemical equation, and show the reaction's yields, reactants' remaining mass, the limiting reagent, and the enthalpy information for the reaction.**
State Dataclass Attributes Minimum of 4 attributes, one of which must be a list/dict of instances of your second dataclass. Briefly describe each attribute, its type, and how it will be meaningfully modified throughout the site.	● balanced_eq (str): The equation the user types on the first page. I only use it to show on the results page, so I never have to parse it. ● reactants (list[Compound]): Every time the user clicks Add reactant, a new Compound gets added here. My loops in calculate go through this list. ● products (list[Compound]): Works the same way as reactants, but for the products. ● lim_reagent (str): calculate sets this to the formula of the limiting reactant. ● enthalpy (float): The heat of the reaction the user enters on the product page. ● endothermic (bool): calculate sets this to True if enthalpy is greater than 0. ● show_work (bool): Set by the checkbox. It decides if the results page shows each step. ● message (str): Holds error messages and warnings, like a bad number or a percent yield over 100.
Second Dataclass Your second dataclass, including its attributes, their types, and how	● formula (str): What the user types in for the compound. ● coefficient (int): The coefficient from the balanced equation.

they will be meaningfully modified (if they are modified)	• molar_mass (float): Found from the formula with ChemPy when the Compound is made. • mass (float): The starting mass for a reactant, or the actual mass for a product. • remaining (float): calculate fills this in for each reactant with the mass left over. • yield_theory (float): calculate fills this in for each product. • yield_percent (float): calculate fills this in for each product. • heat_per_mole (float): calculate fills this in for each product by dividing the heat of the reaction by its coefficient.
Routes (minimum 7) The names of your route functions	index, reactants_page, product_page, results_page, enthalpy_page, save_equation, add_reactant, add_product, calculate, start_over
Pages (minimum 5) A brief description of the pages in your site	• index: Explains what the site does and has a box for the balanced equation and a Start button. • reactants_page: Has boxes for the formula, coefficient, and mass, plus an Add reactant button. It also shows the reactants added so far and has a Done button. • product_page: Has boxes for the product formula, coefficient, actual mass, and the heat of the reaction, plus a checkbox for step-by-step work. It also has an Add product button, shows the products added so far, and has a Calculate button. • results_page: Shows the limiting reactant, the theoretical and percent yield, the leftover masses, whether it's exothermic or endothermic, and the heat per mole. It has buttons to go to the enthalpy page or start over. • enthalpy_page: Explains what the sign of the heat of reaction means and how I got the heat per mole. It has a Back button.

User Input (minimum 3) Description of user input components (TextBox, TextArea, CheckBox, or SelectBox) and their purposes For more information on these input types, see Drafter Documentation.	● A TextBox for the balanced equation. ● TextBoxes for the formula, coefficient, and mass of each reactant. ● TextBoxes for the formula, coefficient, and actual mass of each product, plus the heat of the reaction. ● A CheckBox for showing the step-by-step work.
If Statements (minimum 3) Brief description of what wnill be evaluated in your if statements and how they will meaningfully change the logic of the site	● In add_reactant, add_product, and calculate, I check that every number is filled in and positive. If not, I set message and stay on the same page. ● The Done button only shows up if state.reactants isn't empty. The Calculate button works the same way for state.products. This way the user can't move on with nothing entered. ● Inside the reactant loop, if a reactant has the smallest moles divided by coefficient so far, it becomes the limiting reactant. ● If enthalpy is greater than 0, endothermic is True and the page says endothermic. Otherwise it says exothermic. ● If a percent yield is over 100, I add a warning to check the mass or the equation. ● If show_work is True, the results page also shows each step, from grams to moles to moles divided by coefficient.
Loop (minimum 1) This loop must iterate over the list/dict of dataclass instances in your State attribute. Description of this loop and why it is meaningful to the site	● The first loop goes through state.reactants and finds the moles divided by coefficient for each one to pick the limiting reactant. It has to be a loop because I can't know which one is limiting until I've compared all of them, and the number of reactants changes. ● The second loop goes through state.reactants again and uses the limiting amount to find how

	much of each one is left over. ● The third loop goes through state.products and finds the theoretical yield, percent yield, and heat per mole for each one.

LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

Did not use LLM

Final Submission

When you are finished, download this file as a Word Document (docx) and upload the file to the appropriate assignment on Canvas.